



JERUSALEM COLLEGE OF ENGINEERING
(An Autonomous Institution,
Affiliated to Anna University, Chennai)

QUESTION BANK

JAL2301 DATA SCIENCE

Regulation : 2023

Branch : AI & ML

Year / Semester : II / III

UNIT I**INTRODUCTION**

Course Objective – To understand the data science fundamentals and process.

Course Outcome – Define the data science process.

Bloom's Taxonomy Level - K1, K2.

PART A

S.No.	Question	BTL
1.	What is data science? Give few applications of data science. • Data science is an interdisciplinary field that seeks to extract knowledge or insights from various forms of data. • At its core, data science aims to discover and extract actionable knowledge from data that can be used to make sound business decisions and predictions. • Data science uses advanced analytical theory and various methods such as time series analysis for predicting future. • Applications of data science: anomaly detection, classification, forecasting, pattern detection, recommendation, regression.	K1
2.	What is big data? List few benefits of big data. Big data: Big data is a collection of data sets so large or complex that it becomes difficult to process them using traditional data management techniques such as the RDBMS. Social media, stock market, survey data are few sources of big data. Benefits of big data: 1. Improved customer service 2. Businesses can utilize outside intelligence while taking decisions 3. Reducing maintenance costs 4. Re-develop our products: Big Data can also help us understand how others perceive our products so that we can adapt them or our marketing, if need be. 5. Early identification of risk to the product/services, if any 6. Better operational efficiency	K2
3.	How can the data generated by social media used for data science? Social media is one of the biggest contributors to the big data we have today. For eg., Facebook generates around 500+ terabytes of data every day in the form of content generated by the users like status messages, photos and video uploads, messages, comments, etc. If a company sells a product and uses social media to reach the customer, it can analyze the customer's behavior and preferences from the data collected from social media to decide how to sell the product. It can identify how likely a potential customer will respond to a particular product or service	K2

4.	Define structured data. Structured data is arranged in rows and column format. It helps for application to retrieve and process data easily. Database management system is used for storing structured data. The term structured data refers to data that is identifiable because it is organized in a structure	K1
5.	What is unstructured data? Unstructured data is data that does not follow a specified format. Row and columns are not used for unstructured data. Therefore it is difficult to retrieve required information. Unstructured data has no identifiable structure. Eg. Email.	K2
6.	What is machine - generated data? Give few examples. Machine-generated data is an information that is created without human interaction as a result of a computer process or application activity. This means that data entered manually by an end-user is not recognized to be machine-generated. Eg. web server logs, call detail records, network event logs, and telemetry	K2
7.	Define streaming data. Streaming data is data that is generated continuously by thousands of data sources, which typically send in the data records simultaneously and in small sizes (order of Kilobytes). Eg. “What’s trending” on Twitter, live sporting or music events, and the stock market.	K2
8.	List the stages of data science process. Stages of data science process are as follows: 1. Discovery or Setting the research goal 2. Retrieving data 3. Data preparation 4. Data exploration 5. Data modeling 6. Presentation and automation	K1
9.	What is project charter? All this information about understanding of the business problem is collected in a project charter. It contains A clear research goal • The project mission and context • How you’re going to perform your analysis • What resources you expect to use and where to find them • Proof that it’s an achievable project, or proof of concepts • Deliverables and a measure of success • A timeline	K1
10.	Define the data repositories - data warehouses, data marts and data lakes. Data warehouses contain preprocessed data, collected from various business units.	K2

	A data mart is a subset of the data warehouse and geared toward serving a specific business unit. It contains preprocessed data. Data lakes contain data in its natural or raw format.																			
11.	Mention the advantages of data repositories. Advantages are as follows: i. Data is preserved and archived. ii. Data isolation allows for easier and faster data reporting. iii. Database administrators have easier time tracking problems. iv. There is value to storing and analyzing data.	K2																		
12.	What is data cleansing? Mention few common errors and solutions to fix those errors. Data cleansing is a subprocess of the data science process that focuses on removing errors in the data so that it becomes a true and consistent representation of the processes from where the data originated. <table><tr><th>Common Errors</th><th>Possible Solutions</th></tr><tr><td>Mistakes during data entry</td><td>Manual overrules</td></tr><tr><td>Redundant white space</td><td>Use string functions</td></tr><tr><td>Impossible values</td><td>Manual overrules</td></tr><tr><td>Missing values</td><td>Remove observation or value</td></tr><tr><td>Outliers</td><td>Validate and, if erroneous, treat as missing value</td></tr><tr><td>Deviations from a code book</td><td>Match on keys or else use manual overrules</td></tr><tr><td>Different units of measurement</td><td>Recalculate</td></tr><tr><td>Different levels of aggregation</td><td>Bring to same level of measurement by aggregation or extrapolation</td></tr></table>	Common Errors	Possible Solutions	Mistakes during data entry	Manual overrules	Redundant white space	Use string functions	Impossible values	Manual overrules	Missing values	Remove observation or value	Outliers	Validate and, if erroneous, treat as missing value	Deviations from a code book	Match on keys or else use manual overrules	Different units of measurement	Recalculate	Different levels of aggregation	Bring to same level of measurement by aggregation or extrapolation	K1
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13.	List the data cleaning tasks. Data cleaning are as follows: 1. Data acquisition and metadata 2. Fill in missing values 3. Unified date format 4. Converting nominal to numeric 5. Identify outliers and smooth out noisy data 6. Correct inconsistent data	K2																		
14.	What are outliers and outlier detection? An outlier is an observation that seems to be distant from other observations or that follows a different logic or generative process than the other observations. Outlier detection is the process of detecting and subsequently excluding outliers from a given set of data. The easiest way to find outliers is to use a plot or a table with the minimum and maximum values.	K2																		

15.	What is Euclidean distance? Euclidean distance is used to measure the similarity between observations. It is calculated as the square root of the sum of differences between each point. The Euclidean distance between 2 points (x1, y1) and (x2, y2) = $\sqrt{(x1 - x2)^2 + (y1 - y2)^2}$	K1												
16.	Explain exploratory data analysis. Exploratory Data Analysis (EDA) is a general approach to exploring datasets by means of simple summary statistics and graphic visualizations in order to gain a deeper understanding of data. EDA is used by data scientists to analyze and investigate data sets and summarize their main characteristics, often employing data visualization methods.	K2												
17.	Define data mining. Data mining refers to extracting or mining knowledge from large amounts of data. It is a process of discovering interesting patterns or Knowledge from a large amount of data stored either in databases, data warehouses, or other information repositories.	K1												
18.	What are the steps in models? 1. Selection of a modeling technique and variables to enter in the model 2. Execution of the model 3. Diagnosis and model comparison	K1												
19.	Write about presentation and automation process. After analysing the data successfully and building a well-performing model, they must be presented to the stakeholders. To avoid redoing all of the analysis all the time, the models need to be automated such that the application automatically updates reports, Excel spreadsheets or PowerPoint presentations.	K2												
20.	List few basic statistical descriptions and graphic displays of basic statistical descriptions. <table><tr><th>Basic statistical descriptions</th><th>Graphic displays of basic statistical descriptions</th></tr><tr><td>1. Mean</td><td>1. Scatter plot</td></tr><tr><td>2. Median</td><td>2. Histogram</td></tr><tr><td>3. Mode</td><td>3. Line graph</td></tr><tr><td>4. Variance</td><td>4. Pie chart</td></tr><tr><td>5. Standard Deviation</td><td></td></tr></table>	Basic statistical descriptions	Graphic displays of basic statistical descriptions	1. Mean	1. Scatter plot	2. Median	2. Histogram	3. Mode	3. Line graph	4. Variance	4. Pie chart	5. Standard Deviation		K1
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5. Standard Deviation														

PART B

S. No	Question	BTL
1.	Define data science. Explain the benefits and uses of data science. (16 Marks)	K2
2.	Explain in detail about the facets of data with examples. (16 Marks)	K2
3.	What are the steps in data science process? Briefly explain about any 3 of them with diagram. (16 Marks)	K2
4.	How do you set the research goal and retrieve data? Explain. (16 Marks)	K1
5.	Explain the ways of data preparation step in data science process. (16 Marks)	K2
6.	Explain the process of Exploratory Data Analysis. (16 Marks)	K2
7.	Explain in detail about the different aspects of building models. (16 Marks)	K2
8.	a) List 5 basic statistical descriptions and explain about each of them with relevant examples. (8 Marks)	K2
	b) Explain in detail with examples about graphic displays of basic statistical descriptions. (8 Marks)	K2

UNIT II

DESCRIPTIVE STATISTICS - DESCRIBING DATA

Course Objective – To learn to describe the data for the data science process.

Course Outcome – Understand different types of data description for data science process.

Bloom's Taxonomy Level – K1, K2, K3

PART A

S.No.	Question				BTL
1.	Compare qualitative data and quantitative data.				K2
	Qualitative Data		Quantitative Data		
	Qualitative data (or categorical data) provides information about the quality of an object or information which cannot be measured.		Quantitative data is the one that focuses on numbers and mathematical calculations and can be calculated and computed.		
	Qualitative data cannot be expressed as a number. They represent nominal scales		Quantitative data can be expressed as a number or quantified.		
	Examples of qualitative data are gender, economic status and religious preference.		Examples of quantitative data are scores on achievement tests, number of hours of study or weight of a subject.		
2.	Classify the below list of data into their types:				K3
	(a) ethnic group, (b) age, (c) family size, (d) academic major, (e) sexual preference, (f) IQ score, (g) net worth (dollars), (h) third-place finish, (i) gender, (j) temperature.				
	(a) Ethnic group	Qualitative data	(f) IQ score	Quantitative data	
	(b) Age	Quantitative data	(g) Net worth (dollars)	Quantitative data	
	(c) Family size	Quantitative data	(h) Third-place finish	Ranked data	
	(d) Academic major	Qualitative data	(i) Gender	Qualitative data	
	(e) Sexual preference	Qualitative data	(j) Temperature	Quantitative data	

3.	What are variables? Differentiate discrete and continuous variables. A variable is a characteristic or property that can take on different values. A <i>discrete variable</i> consists of isolated numbers separated by gaps. Examples include most counts, such as the number of children in a family (1, 2, 3, etc., but never 1½), the number of foreign countries you have visited; and the current size of the Indian population. A <i>continuous variable</i> consists of numbers whose values, at least in theory, have no restrictions. Examples include amounts, such as weights of male statistics students, durations, such as the reaction times of grade school children to a fire alarm; and standardized test scores, such as those on the Scholastic Aptitude Test	K2												
4.	Differentiate independent and dependent variables. <table><thead><tr><th>Independent Variable</th><th>Dependent Variable</th></tr></thead><tbody><tr><td>An Independent variable is a variable whose value never depends on another variable but the researcher.</td><td>A dependent variable is a variable whose value depends on another variable.</td></tr><tr><td>The independent variable is the presumed cause.</td><td>The dependent variable is the presumed effect.</td></tr><tr><td>Any change in the dependent variable does not affect the independent variable.</td><td>Any change in the independent variable also affects the dependent variable.</td></tr><tr><td>Independent variables are the predictors or regressors.</td><td>Dependent variables are often referred as the predicted variable.</td></tr><tr><td>Eg. Intelligence (IQ level)</td><td>Eg. Test score</td></tr></tbody></table>	Independent Variable	Dependent Variable	An Independent variable is a variable whose value never depends on another variable but the researcher.	A dependent variable is a variable whose value depends on another variable.	The independent variable is the presumed cause.	The dependent variable is the presumed effect.	Any change in the dependent variable does not affect the independent variable.	Any change in the independent variable also affects the dependent variable.	Independent variables are the predictors or regressors.	Dependent variables are often referred as the predicted variable.	Eg. Intelligence (IQ level)	Eg. Test score	K2
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Eg. Intelligence (IQ level)	Eg. Test score													
5.	What is Confounding Variable? Give example. An uncontrolled variable that compromises the interpretation of a study is known as a confounding variable. For example, if we have an association between two variables (X and Y), and that association is due entirely to the fact that both X and Y are affected by a third variable (Z), then we would say that the association between X and Y is spurious and that it is a result of the effect of a confounding variable (Z). Age, sex, ethnicity, education, diet, exercise, smoking, BMI... (Confounding variables) Z X Alcohol consumption (Independent variable) Y Mortality (Dependent variable)	K2												
6.	What is frequency distribution? A frequency distribution is a collection of observations produced by sorting observations into classes and showing their frequency (f) of occurrence in each class.	K1												
7.	What are relative and cumulative frequencies? Relative frequency distributions show the frequency of each class as a part or fraction of the total frequency for the entire distribution.	K1												

	Cumulative frequency distributions show the total number of observations in each class and in all lower-ranked classes.	
8.	What are Stem and Leaf displays? Stem and Leaf display is a technique for summarizing quantitative data. Stem and Leaf displays allow to display raw data visually. They are ideal for summarizing distributions, such as that for weight data, without destroying the identities of individual observations. Each raw score is divided into a stem and a leaf. The leaf is typically the last digit of the raw value. The stem is the remaining digits of the raw value. Data points are split into a leaf (usually the ones digit) and a stem (the other digits).	K2
9.	What do the typical shapes for smoothed frequency polygons represent? a) Normal shape – A bell shaped distribution is a true symmetric distribution of observed values. b) Bimodal shape – It reflects the coexistence of two different types of observations in the same distribution. c) Positively skewed shape - A lopsided distribution caused by a few extreme observations in the positive direction (to the right of the majority of observations). d) Negatively skewed shape - A lopsided distribution caused by a few extreme observations in the negative direction (to the left of the majority of observations).	K2
10.	Which average can be used to describe the central tendency of the distribution (a) if distribution is not skewed and (b) if distribution is skewed? (a) If distribution is not skewed - the values of the mode, median, and mean are similar, and any of them can be used to describe the central tendency of the distribution. (b) If distribution is skewed – since the values of the three averages can differ appreciably, the mean describes the central tendency of a distribution.	K2
11.	How is the median rank for ranked data identified? When the data consist of a series of ranks, with its ordinal level of measurement, the median rank can be obtained by the middlemost or average of the two middlemost ranks.	K2
12.	What are measures of variability? List the different measures of variability. What is the goal of variability? Measures of variability means the measures of the amount by which scores are dispersed or scattered in a distribution. The different measures of variability are the range, the interquartile range, the variance, and most important, the standard deviation. The goal for variability is to obtain a measure of how spread out the scores are in a distribution. A measure of variability usually accompanies a measure of central tendency as basic descriptive statistics for a set of scores.	K1
13.	What is range? What are the limitations of range? The range is the difference between the largest and smallest scores. Range = Maximum value - Minimum value <i>Limitations of range:</i> • Since its value depends on only two scores - the largest and the smallest - it fails to use the information provided by the remaining scores.	K2

	The value of the range tends to increase with increases in the total number of scores.	
14.	What is sum of squares? The sum of squared deviation scores, or the sum of squares (SS), is an important measure because it's a major component in calculations for the variance, as well as many other statistical measures. The two formulas for the sum of squares are the definition formula and the computation formula. Definition formula: $SS = \sum (X - \mu)^2$ Computation formula: $SS = \sum X^2 - (\sum X)^2 / N$	K1
15.	Give the formula for variance and standard deviation for population and sample. For population: Variance for population, $\sigma^2 = \frac{\text{Sum of all squared deviation scores}}{\text{Number of scores}} = \frac{SS}{N}$ Standard deviation for population $= \sigma = \sqrt{\sigma^2}$ For sample: Variance for sample, $s^2 = \frac{SS}{n-1}$ Standard deviation for sample $= s = \sqrt{s^2}$	K1
16.	What is meant by degrees of freedom? Degrees of freedom (df) refers to the number of values that are free to vary, given one or more mathematical restrictions, in a sample being used to estimate a population characteristic.	K2
17.	Point out the properties of the Normal Curve. - The normal curve is a theoretical curve obtained from a mathematical equation. It is defined for a continuous variable and noted for its symmetrical bell-shaped form. - Because the normal curve is symmetrical, its lower half is the mirror image of its upper half. - Being bell shaped, the normal curve peaks above a point midway along the horizontal spread and then tapers off gradually in either direction from the peak (without actually touching the horizontal axis, since, in theory, the tails of a normal curve extend infinitely far). - The values of the mean, median (or 50th percentile), and mode, located at a point midway along the horizontal spread, are the same for the normal curve.	K2
18.	Write a short notes on z-score. A z score is a unit-free, standardized score that, regardless of the original units of measurement, indicates how many standard deviations a score is above or below the mean of its distribution. Z score, $z = \frac{X - \mu}{\sigma}$ A z score consists of two parts: 1. A positive or negative sign indicating whether it's above or below the mean; and 2. A number indicating the size of its deviation from the mean in standard deviation units.	K2

19.	How are normal curve problems solved? Concentrate on the logic of the solution, using rough graphs of normal curves as an aid to visualize the solution. Only after thinking through to a solution should you do any calculations and consult the normal tables.	K2
20.	Give the stepwise procedure for finding proportions for One Score. 1. Sketch a normal curve and shade in the target area. 2. Plan your solution according to the normal table. 3. Convert X to z . 4. Find the target area.	K1

PART B

S. No.	Question	BTL																																	
1.	Write short notes on different types of data. (8 Marks)	K2																																	
2.	What are the different types of variables used in data analysis? (16 Marks) Demonstrate each of them with relevant examples.	K2																																	
3.	Describe the following in detail with relevant examples: (16 Marks) (a) Frequency distributions for quantitative data (b) Relative frequency distributions (c) Cumulative frequency distributions (d) Frequency distributions for qualitative (nominal) data	K2																																	
4.	The number of friends reported by Facebook users (16 Marks) is summarized in the following frequency distribution: <table><tr><td>FRIENDS</td><td>f</td><td>a) What is the shape of this distribution?</td></tr><tr><td>400 – above</td><td>2</td><td>b) Find the relative frequencies.</td></tr><tr><td>350 – 399</td><td>5</td><td>c) Find the cumulative frequencies.</td></tr><tr><td>300 – 349</td><td>12</td><td>d) Find the approximate percentile rank of the interval 300–349.</td></tr><tr><td>250 – 299</td><td>17</td><td>e) Convert to a histogram.</td></tr><tr><td>200 – 249</td><td>23</td><td>f) Why would it not be possible to convert to a stem and leaf display?</td></tr><tr><td>150 – 199</td><td>49</td><td></td></tr><tr><td>100 – 149</td><td>27</td><td></td></tr><tr><td>50 – 99</td><td>29</td><td></td></tr><tr><td>0 – 49</td><td>36</td><td></td></tr><tr><td>Total</td><td>200</td><td></td></tr></table>	FRIENDS	f	a) What is the shape of this distribution?	400 – above	2	b) Find the relative frequencies.	350 – 399	5	c) Find the cumulative frequencies.	300 – 349	12	d) Find the approximate percentile rank of the interval 300–349.	250 – 299	17	e) Convert to a histogram.	200 – 249	23	f) Why would it not be possible to convert to a stem and leaf display?	150 – 199	49		100 – 149	27		50 – 99	29		0 – 49	36		Total	200		K3
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5.	The IQ scores for a group of 35 high school dropouts are as follows:									K3	
	91	85	84	79	80	87	96	75	86		104
	95	71	105	90	77	123	80	100	93		108
	98	69	99	95	90	110	109	94	100		103
	112	90	90	98	89						
	i. Construct a frequency distribution for grouped data ii. Relative Frequency distribution iii. Cumulative Frequency distribution										
6.	Explain some of the most common types of graphs and their (16 Marks) shapes used to describe quantitative and qualitative data.									K1	
7.	Explain how data can be described with averages. (16 Marks)									K2	
8.	What is the importance of variability? Describe about the (16 Marks) following measures of variability: (a) Range (b) Variance (c) Standard deviation (d) Sum of squares (e) Interquartile range (IQR)									K2	
9.	(i) Write in detail about normal curve and solving normal curve (8 Marks) problems. (ii) Give examples and explanation for (a) Finding proportions for one score (b) Finding Proportions between Two Scores (c) Finding Proportions beyond Two Scores									K2	
10.	(i) Write short notes on z scores. (8 Marks)									K2	
	(ii) The body mass index (BMI) measures body size in people by dividing weight (in pounds) by the square of height (in inches) and then multiplying by a factor of 703. A BMI less than 18.5 is defined as underweight; between 18.5 and 24.9 is normal; between 25 and 29.9 is overweight; and 30 or more is obese. It is well established that Americans have become heavier during the last half century. Assume that the positively skewed distribution of BMIs for adult American males has a mean of 28 with a standard deviation of 4. (8 Marks)										
	a) Would the median BMI score exceed, equal, or be exceeded by the mean BMI score of 28? b) What z score defines overweight? c) What z score defines obese?										
										K3	

UNIT III

DESCRIPTIVE STATISTICS - DESCRIBING RELATIONSHIPS

Course Objective – To learn to describe the relationship between data.

Course Outcome – Gain knowledge on relationships between data.

Bloom's Taxonomy Level – K1, K2, K3

PART A

S.No.	Question	BTL
1.	What are positive and negative relationships? - Insofar as relatively low values are paired with relatively low values, and relatively high values are paired with relatively high values, the relationship is positive - "You get what you give." - Insofar as relatively low values are paired with relatively high values, and relatively high values are paired with relatively low values, the relationship is negative - "You get the opposite of what you give."	K2
2.	Consider Helen sent 10 greeting cards to her friends and she received back 8 cards. What is the kind of relationship it is? Brief on it. Negative relationship. As the number of greeting cards sent is higher and the number of greeting cards received is lower – "You get the opposite of what you give" represents negative relationship..	K3
3.	Indicate whether the following statements suggest a positive or negative relationship: (a) More densely populated areas have higher crime rates. (b) School children who often watch TV perform more poorly on academic achievement tests. (c) Heavier automobiles yield poorer gas mileage. (d) Better-educated people have higher incomes. (e) More anxious people voluntarily spend more time performing a simple repetitive task. (a) Positive. The crime rate is higher, square mile by square mile, in densely populated cities than in sparsely populated rural areas. (b) Negative. As TV viewing increases, performance on academic achievement tests tends to decline. (c) Negative. Increases in car weight are accompanied by decreases in miles per gallon. (d) Positive. Increases in educational level—grade school, high school, college—tend to be associated with increases in income. (e) Positive. Highly anxious people willingly spend more time performing a simple repetitive task than do less anxious people.	K3
4.	What is a scatterplot? A scatterplot is a graph containing a cluster of dots that represents all pairs of scores. Any dot cluster can be used as a preview of a fully measured relationship.	K1
5.	How does a dot cluster reflect relationships? A dot cluster that has a slope from the lower left to the upper right, reflects a positive relationship.	K2

	• A dot cluster that has a slope from the upper left to the lower right, reflects a negative relationship. • A dot cluster that lacks any apparent slope, reflects little or no relationship.									
6.	Differentiate perfect and curvilinear relationships. <table><thead><tr><th>Perfect Relationship</th><th>Curvilinear Relationship</th></tr></thead><tbody><tr><td>A dot cluster that equals (rather than merely approximates) a straight line reflects a perfect relationship between two variables.</td><td>A dot cluster approximates a bent or curved line, reflects a curvilinear relationship.</td></tr><tr><td>Simpler</td><td>More complex</td></tr><tr><td>In practice, perfect relationships are most unlikely.</td><td>Most of the relationships in practice are curvilinear.</td></tr></tbody></table>	Perfect Relationship	Curvilinear Relationship	A dot cluster that equals (rather than merely approximates) a straight line reflects a perfect relationship between two variables.	A dot cluster approximates a bent or curved line, reflects a curvilinear relationship.	Simpler	More complex	In practice, perfect relationships are most unlikely.	Most of the relationships in practice are curvilinear.	K3
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Simpler	More complex									
In practice, perfect relationships are most unlikely.	Most of the relationships in practice are curvilinear.									
7.	What is a correlation coefficient? What are its properties? A correlation coefficient r is a number between -1 and 1 that describes the relationship between pairs of variables. The properties of correlation coefficient r: 1. The sign of r indicates the type of linear relationship, whether positive or negative. 2. The numerical value of r, without regard to sign, indicates the strength of the linear relationship.	K2								
8.	Write in brief about how r is independent of units of measurement? The value of r is independent of the original units of measurement. In fact, the same value of r describes the correlation between height and weight for a group of adults, regardless of whether height is measured in inches or centimeters or whether weight is measured in pounds or grams. In effect, the value of r depends only on the pattern among pairs of scores, which in turn show no traces of the units of measurement for the original X and Y scores.	K2								
9.	Mention the purpose of verbal descriptions. When interpreting a brand new r, it will be helpful to translate the numerical value of r into a verbal description of the relationship. For eg. an r of $.70$ for the height and weight of college students could be translated into “Taller students tend to weigh more”. If there is trouble verbalizing the value of r, the original scatterplot can be referred back or, if necessary, a rough scatterplot corresponding to the value of r can be visualized. Any detectable dot cluster can be used to think through the relationship based on whether it has a slope from the lower left to the upper right.	K2								
10.	Write the computation formula for correlation coefficient. Correlation coefficient, $r = \frac{SP_{xy}}{\sqrt{SS_x SS_y}}$ Where $SS_x = \sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$ $SS_y = \sum (Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$ $SP_{xy} = \sum (X - \bar{X})(Y - \bar{Y}) = \sum XY - \frac{(\sum X)(\sum Y)}{n}$	K1								

11.	Give and explain the Least Squares Regression Equation. If a linear relationship exists between two variables, then one variable can be predicted from the other by using the least squares regression equation. The exact least squares regression line for any scatterplot is given by the following equation. $Y' = b X + a$ where Y' represents the predicted value, X represents the known value and b and a represent numbers calculated from the original correlation analysis, where $b = r \sqrt{\frac{SS_y}{SS_x}}$ where r represents the correlation between X and Y and $a = \bar{Y} - b \bar{X}$	K2								
12.	Define standard error of estimate. The standard error of estimate is a special kind of standard deviation that roughly reflects the average amount of predictive error. The value of the standard error of estimate depends mainly on the size of the correlation coefficient. Standard error of estimate, = $s_{y x} = \sqrt{\frac{SS_y (1-r^2)}{n-2}}$	K1								
13.	Assume that an r of .30 describes the relationship between educational level (highest grade completed) and estimated number of hours spent reading each week. More specifically: <table><tr><td>EDUCATIONAL LEVEL (X)</td><td>WEEKLY READING TIME (Y)</td></tr><tr><td>$\bar{X} = 13$</td><td>$\bar{Y} = 8$</td></tr><tr><td>$SS_x = 25$</td><td>$SS_y = 50$</td></tr><tr><td colspan="2">$r = .30$</td></tr></table> (a) Determine the least squares equation for predicting weekly reading time from the given educational level - Faith's education level is 15. What is her predicted reading time? (b) Calculate the standard error of estimate for the data, assuming that the correlation of .30 is based on $n = 35$ pairs of observations. (c) Supply a rough interpretation of the standard error of estimate (a) $b = r \sqrt{\frac{SS_y}{SS_x}} = (0.30) \sqrt{\frac{50}{25}} = 0.42$ $a = \bar{Y} - b \bar{X} = 8 - (0.42) 13 = 2.54$ Faith's predicted reading time $Y' = b X + a = (0.42) 15 + 2.54 = 8.84$ (b) $s_{y x} = \sqrt{\frac{SS_y (1-r^2)}{n-2}} = \sqrt{\frac{50(1-0.30)^2}{35-2}} = \sqrt{\frac{50(1-0.9)}{33}} = \sqrt{1.38} = 1.17$ (c) Roughly indicates the average amount by which the prediction is in error.	EDUCATIONAL LEVEL (X)	WEEKLY READING TIME (Y)	$\bar{X} = 13$	$\bar{Y} = 8$	$SS_x = 25$	$SS_y = 50$	$r = .30$		K3
EDUCATIONAL LEVEL (X)	WEEKLY READING TIME (Y)									
$\bar{X} = 13$	$\bar{Y} = 8$									
$SS_x = 25$	$SS_y = 50$									
$r = .30$										

14.	What is meant by interpretation of r^2? The squared correlation coefficient, r^2, provides us with a key interpretation of the correlation coefficient and a measure of predictive accuracy that supplements the standard error of estimate, $S_{y x}$. It indicates the proportion of total variability in one variable that is predictable from its relationship with the other variable. $r^2 = \frac{SS_{Y'}}{SS_Y} = \frac{SS_Y - SS_{Y X}}{SS_Y} = \frac{\sum(Y' - \bar{Y})^2}{SS_Y}$	K2
15.	Define error variability. Error variability (or) sum of squares: To more precisely evaluate the accuracy of two predictive efforts, sum of squares is used as a measure of the collective errors produced by each effort. The sum of squares of any set of deviations is called as errors. It can be calculated by first squaring each error (to eliminate negative signs), then summing all squared errors. The error variability for the repetitive prediction of the mean can be designated as SS_y, where $SS_y = \sum(Y - b\bar{Y})^2$	K1
16.	What is gain in accuracy? Gain in accuracy = $SS_y - SS_{y x}$ Proportion or percent gain in predictive accuracy = $\frac{SS_y - SS_{y x}}{SS_y}$	K1
17.	Write a note on multiple regression equations. When several X, or <i>predictor variables</i> are involve, a more complex equation is required. For instance, to predict college GPA might use the following equation: $Y' = 0.410(X1) + 0.005(X2) + 0.001(X3) + 1.03$ where Y' represents predicted college GPA and $X1$, $X2$, and $X3$ refer to high school GPA, IQ score, and SAT score, respectively. By using the combined predictive power of several predictor variables, these multiple regression equations supply more accurate predictions for Y' than obtaining from a simple regression equation.	K2
18.	What is regression toward the mean? Regression toward the mean refers to a tendency for scores, particularly extreme scores, to shrink toward the mean.	K1
19.	How can regression fallacy be avoided? The regression fallacy is committed whenever regression toward the mean is interpreted as a real, rather than a chance, effect. It can be avoided by splitting the subset of extreme observations into two groups. Any observed difference between the two groups would be viewed as a real difference not attributable to the regression effect.	K2
20.	After a group of college students attended a stress-reduction clinic, declines were observed in the anxiety scores of those who, prior to attending the clinic, had scored high on a test for anxiety. (a) Can this decline be attributed to the stress-reduction clinic? Explain your answer. (b) What type of study, if any, would permit valid conclusions about the effect of the stress reduction clinic?	K3

	(a) No, because the observed decline could be due to regression toward the mean, given that the students scored high on the anxiety test prior to attending the clinic. (b) An experiment where students who score high on the anxiety test are randomly assigned either to attend the stress-reduction clinic or to be in a control group.	
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PART B

S. No.	Question	BTL																		
1.	Categorize the different types of relationships interpreted using scatter plots. (16 Marks)	K2																		
2.	Explain in detail about correlation coefficient for quantitative data. (16 Marks)	K2																		
3.	How can correlation coefficient be calculated? Give and explain about the formula used. Apply the same for the following data: <table><thead><tr><th>NAME</th><th>CARDS SENT</th><th>CARDS RECEIVED</th></tr></thead><tbody><tr><td>Steve</td><td>13</td><td>14</td></tr><tr><td>Donald</td><td>9</td><td>18</td></tr><tr><td>Andrea</td><td>7</td><td>12</td></tr><tr><td>John</td><td>5</td><td>10</td></tr><tr><td>Rohan</td><td>1</td><td>6</td></tr></tbody></table> (16 Marks)	NAME	CARDS SENT	CARDS RECEIVED	Steve	13	14	Donald	9	18	Andrea	7	12	John	5	10	Rohan	1	6	K3
NAME	CARDS SENT	CARDS RECEIVED																		
Steve	13	14																		
Donald	9	18																		
Andrea	7	12																		
John	5	10																		
Rohan	1	6																		
4.	Write in detail about regression line and least squares regression line. (16 Marks)	K1																		
5.	What is meant by standard error of estimate? How is it calculated? (16 Marks)	K2																		
6.	Write a note on r^2 and its interpretation. (8 Marks)	K2																		
7.	Each of the following pairs represents the number of licensed drivers (X) and the number of cars (Y) for seven houses in my neighborhood: <table><thead><tr><th>DRIVERS (X)</th><th>CARS (Y)</th></tr></thead><tbody><tr><td>5</td><td>4</td></tr><tr><td>5</td><td>3</td></tr><tr><td>2</td><td>2</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>2</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr></tbody></table> a) Construct a scatterplot to verify a lack of pronounced curvilinearity. b) Determine the least squares equation for these data. (Remember, you will first have to calculate r, SS_y and SS_x) c) Determine the standard error of estimate, $s_{y x}$, given that $n = 7$. (8 Marks)	DRIVERS (X)	CARS (Y)	5	4	5	3	2	2	2	2	3	2	1	1	2	2	K3		
DRIVERS (X)	CARS (Y)																			
5	4																			
5	3																			
2	2																			
2	2																			
3	2																			
1	1																			
2	2																			

	d) Predict the number of cars for each of two new families with two and five drivers.	
8.	In studies dating back over 100 years, it's well established that regression toward the mean occurs between the heights of fathers and the heights of their adult sons. Indicate whether the following statements are true or false. a) Sons of tall fathers will tend to be shorter than their fathers. b) Sons of short fathers will tend to be taller than the mean for all sons. c) Every son of a tall father will be shorter than his father. d) Taken as a group, adult sons are shorter than their fathers. e) Fathers of tall sons will tend to be taller than their sons. f) Fathers of short sons will tend to be taller than their sons but shorter than the mean for all fathers. (8 Marks)	K3